

Mars Habitat

Campaign 1 - Case 03 - Arcadia Planitia, Mars

SOLAR
AGRICULTURAL
AGENCY

Name _____ Date _____ Period _____

Mission question

How can the habitat receive an adequate total amount of light while potato plants still fail to form normal green new growth?

CREW

New growth bleaches first. Added iron and nitrogen did not help.

SENSORS

PPFD is 280 $\mu\text{mol m}^{-2} \text{s}^{-1}$. Light travels through a 12 m pipe plus LEDs.

PLANTS

Old leaves retain green. Roots are healthy.

LOGS

No spectral test was done before the failure.

TASK 1 - Define the measurement

Complete both statements.

PPFD tells us:

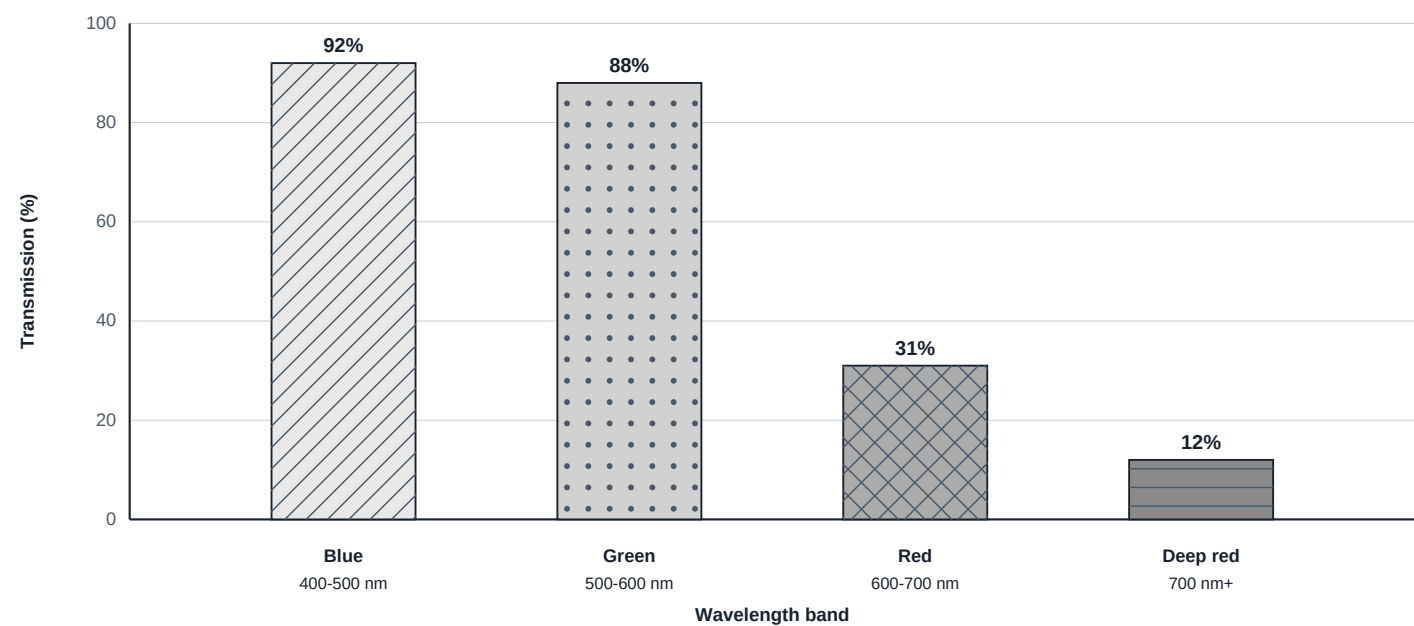
PPFD does not tell us:

TASK 2 - Read the spectral-transmission data

Use only the four values shown. Which band inside 400-700 nm has the lowest transmission?

Band	Range	Transmission
Blue	400-500 nm	92%
Green	500-600 nm	88%
Red	600-700 nm	31%
Deep red	700 nm+	12%

Surface intake to habitat output transmission

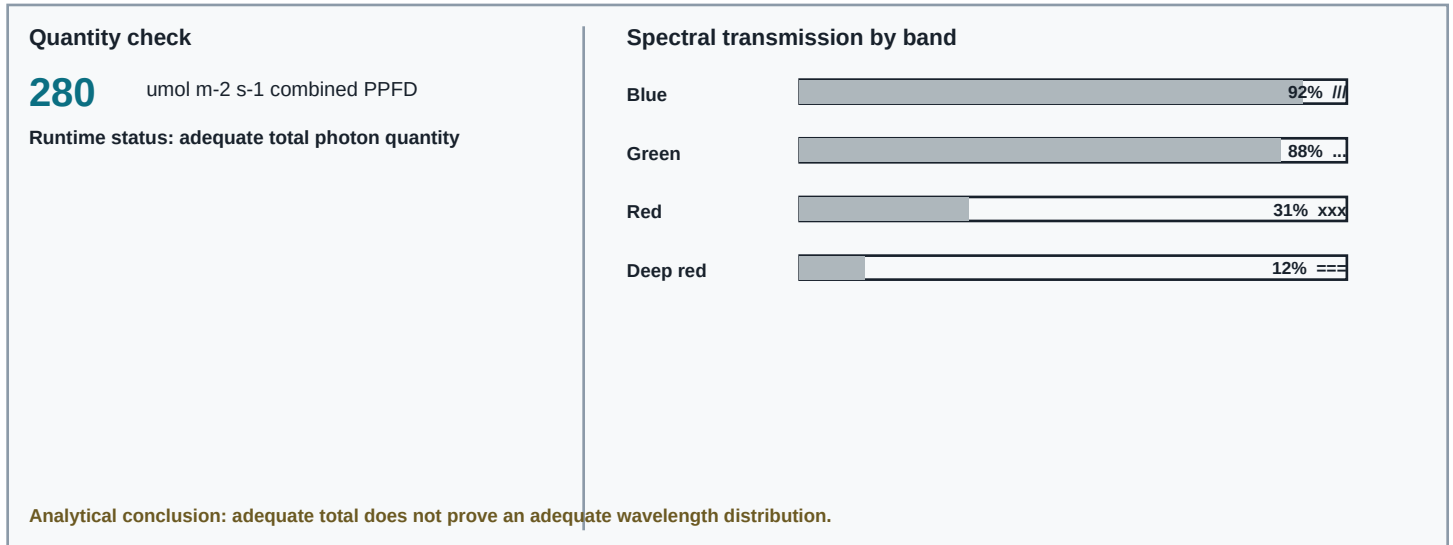


SOURCE STATUS: These four whole-number transmission values are game-provided.

LOWEST IN-PAR TRANSMISSION AND HOW YOU KNOW

TASK 3 - Compare quantity and quality

The total PPFD is 280 $\mu\text{mol m}^{-2} \text{s}^{-1}$ and the game describes it as adequate. Compare that quantity with the uneven distribution below.



Why does the explanation “not enough total light” conflict with the sensor reading?

What is wrong with the delivered light instead?

TASK 4 - Connect the symptom pattern

Old leaves retain green. New leaves become pale first. Roots are healthy. Nutrient additions did not help.

Observation	Meaning
Old tissue remains green	Existing chlorophyll is still present.
New tissue bleaches first	New chlorophyll formation is disrupted.
Healthy roots	Root poisoning is not supported.

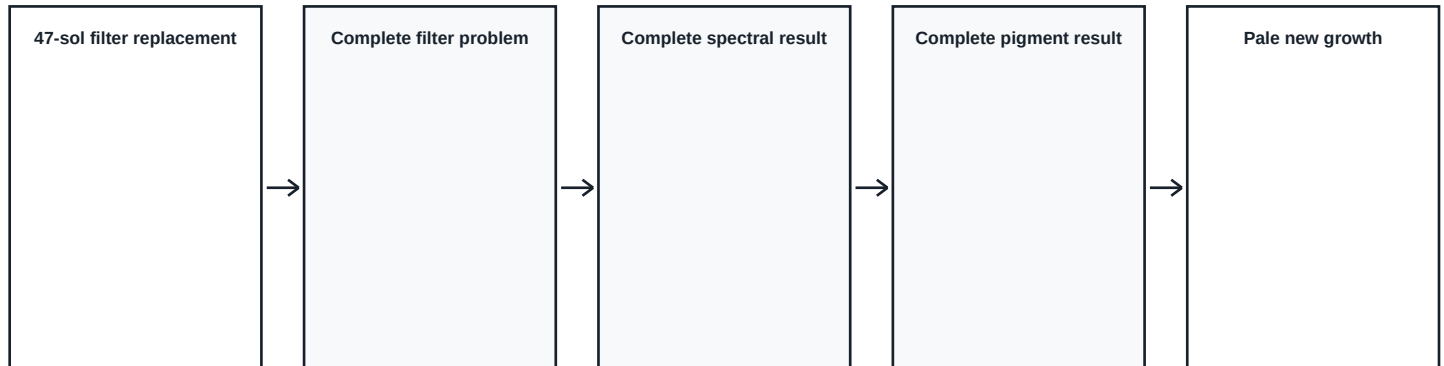
EXPLAIN THE PATTERN

TASK 5 - Select and reject diagnoses

Choose the diagnosis that fits all evidence. Then reject one other option.

TASK 6 - Model the mechanism

Complete the causal chain from the light-delivery system to the visible symptom.



Case mechanism model. General plant-light responses also involve blue, green, and red wavelengths.

EXPLAIN THE CHAIN IN WORDS

SCIENCE BOUNDARY: The case focuses on red-band rejection. Real plants also respond to blue, green, and other red wavelengths.

TASK 7 - Write the case conclusion

Write the claim and evidence. Reasoning continues on the next page.

Claim

Evidence

TASK 7 - Write the case conclusion - continued

Reasoning: connect the data to the mechanism and symptom pattern.

TASK 8 - Transfer the analysis

Why might making every wavelength brighter be the wrong first fix? What should the analyst measure next?

TASK 9 - Exit ticket

Name the first two lighting measurements you would compare in a new case and explain why both are needed.